

CSE331: Automata and Computability
Assignment 02

Solve all the questions given below:

- 1) List five use cases for applying the knowledge from our CSE331 course. Describe in detail.
- 2) Proof L is a non-regular language using the Pumping Lemma:
[If you are not seeing any Qs, then View in the light theme]

a) $L = \{w \in \{a, b\}^* \mid w = a^i b^j, i < j, i \geq 0\}$

b) $L = \{0^n 1^m \mid n = 3q + r_1, m = 3q + r_2, q \geq 0, 0 \leq r_1, r_2 \leq 2\}$

c) $L = \{w \in \{0, 1\}^* \mid w = 0^n 1, n \geq 0\}$

d) $L = \{w \in \{0, 1\}^* \mid w = 10^{n^2}, n \geq 0\}$

e) $L = \{w \in \{0, 1\}^* \mid w = 0^a 1^b 1^c 0^d, a + b = c + d, a, b, c, d \geq 0\}$

f) $L = \{w \in \Sigma^* \mid w = a^n 1^m 0^k, \text{ where } n = m \text{ or } n \neq k\}$

g) $L = \{w \in \{0, 1, 2\}^* \mid w = 0^n 1^m 2^{n-m}, n \geq m \geq 0\}$

h) ~~$L = \{w \in \{a, b\}^* \mid w = a^i b^j, \text{ where } i > j, \text{ and } j \geq 0\}$~~

3) Design a Context Free Grammar for the Language:

a) $L = \{w \in \{a, b, c, p, q, r, \#\}^* \mid w = a^i \#^n c^j p^{2x} q^y r^z b^k \text{ where } i=j+k, y=3x+z, z \text{ is even, } n \text{ is odd, and } i, j, k, n, x, y, z \geq 0\}$

b) $L = \{w \in \{0, 1, 2\}^* \mid w = 0^i 2^j 1^k, [\text{whereconditions.....}]\}$

where...

- i) $i = k, i, k \geq 1 \text{ and } j \geq 2$
- ii) $i = 3k, j \text{ is odd, and } i, j, k \geq 0$

- iii) i is a multiple of two, k is two more than a multiple of 3, $j = k+i$, and $i, j, k \geq 0$
- iv) $i+j > k$ and $i, j, k \geq 0$
- v) $i+k$ is even, $j = i+k$ and $j \geq 1$
- c) $L = \{w \in \{0,1\}^*: \text{ the parity of 0s and 1s is different in } w\}$
- d) $L = \{w \in \{0,1\}^*: \text{ the number of 0s and 1s are different in } w\}$
 [Hint: First, try to solve for an equal number of 0s and 1s in w]
- e) $L = \{w \in \{a, b\}^*: w \text{ is a palindrome}\}$. Design a CFG for $\overline{L}$ (Complement of L)
- f) $L = \{w \in \{a, b\}^*: w \text{ does not contain any palindromic substring of length two}\}$
- g) $L = \{w\#x \mid w^R \text{ is a substring of } x \text{ where } w \in \{0, 1\}^*\}$.
 w^R Denotes reverse of string, w .
- h) $L = \{1^i 0 2^j 1^k \mid i, j, k \geq 0, 3i \geq 4k + 2, j \text{ is not divisible by three}\}$
- i) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$
 $L1 = \{w \in \Sigma^*: w \text{ contains exactly two 1s}\}$
 $L2 = \{x\#y : x \in \Sigma^*, y \in L1, |x| = |y|\}$
 Construct a CFG for $L2$.
- j) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$
 $L1 = \{w \in \Sigma^*: w \text{ contains at least three 1s}\}$
 $L2 = \{x\#y : x \in (\Sigma\Sigma)^*, y \in L1, |x| = |y|\}$
 Construct a CFG for $L2$.
- k) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{a,b\}$
 $L1 = \{w \in \Sigma^*: w \text{ contains } bb \text{ as a substring}\}$
 $L2 = \{w1\#w2 : w1, w2 \in L1 \text{ and } |w1| = |w2|\}$
 Construct a CFG for $L2$.

- 4) For all the problems in Question(3), now construct the Pushdown Automata (PDA).

5) Derivation, Parse Tree, Ambiguity

- i) Given the Context-Free Grammar, answer the following questions

$$A \rightarrow A1 \mid 0A1 \mid 01$$

- (a) Give a leftmost derivation for the string 001111.
- (b) Sketch the parse tree corresponding to the derivation you gave in (a).
- (c) Demonstrate that there are two more parse trees (apart from the one you already found in (b)) for the same string.
- (d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)

- ii) Given the Context-Free Grammar, answer the following questions:

$$\begin{aligned} A &\rightarrow 1A \mid 1C \mid 0B \mid 00A \\ B &\rightarrow 0A \mid 1B \mid 00B \\ C &\rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \varepsilon \end{aligned}$$

- (a) Give a leftmost derivation for the string 01011001. (3 points)
- (b) Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)
- (c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
- (d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)
- (e) Design an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

6) Turing Machine

- a) $L = \{1^{3^n} \mid n \geq 0\}$
- b) $L = \{w \in \{0,1,2\}^* : 0^i 1^j 2^k \text{ where } i=j=k \text{ and } i,j,k \geq 0\}$
- c) $L = \{w \in \{0,1,2\}^* : 0^i \# 1^j \# 2^k \text{ where } i*j=k \text{ and } i,j,k \geq 0\}$ [Idea described in Sipser's Book]
- d) $L = \{w \in \{a,b\}^* : w \text{ is an odd length palindrome}\}$
- e) $L = \{w \# w^R \# w \mid w \in \{a,b\}^*\}$ [w^R means reverse of w]

- f) Let $\text{HALT_TM} = \{ \langle M, w \rangle : M \text{ is a TM and } M \text{ halts on input } w \}$.
 Given that HALT_TM is undecidable, prove that the collection of Turing-recognizable languages is not closed under complement.

CSE331: Automata and Computability
Practice Sheet
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Question 1: Designing Context-Free Grammar

Give Context-free Grammar that generates the language

- a) $L = \{w \in \{0,1\}^* : w \text{ starts with '0' and the length of } w \text{ is even.}\}$
- b) $L = \{w \in \{a,b\}^* : \text{every second letter in } w \text{ is a 'b'.}\}$
- c) $L = \{w \in \{a,b\}^* : \text{the length of } w \text{ is divisible by three}\}$
- d) $L = \{w \in \{0,1\}^* : w \text{ starts with and ends with a different symbol.}\}$
- e) $L = \{w \in \{a,b\}^* : \text{the number of 'a' is at least the number of 'b' in } w.\}$
- f) $L = \{w \in \{a,b\}^* : \text{the length of } w \text{ is two more than a multiple of six}\}$
- g) $L = \{w \in \{0,1\}^* : w = 0^n 1^{2n+3}, n \geq 0 \}$
- h) $L = \{w \in \{0,1\}^* : w = 0^n 1^n, \text{ where } n \text{ is odd.}\}$
- i) $L = \{w \in \{0,1,2\}^* : w = 0^i 1^j 2^k \text{ where } j \geq 2i + 3k, \text{ and } i, k \geq 0\}$
- j) $L = \{w \in \{0,1,2,3\}^* : w = 0^i 1^j 2^k 3^m \text{ where } i=m, j \geq 3k+2, \text{ and } m, k \geq 0\}$
- k) $L = \{w \in \{0,1,2\}^* : w = 0^i 1^j 2^k \text{ where } i > 2j + 3k, \text{ and } j, k \geq 0\}$
- l) If $A = \{w \in \{0,1\}^* : w \text{ contains at least two 0s}\}$, then construct $L = \{w \in \{0,1\}^* : w = 0^{3i} v 1^{2i} \text{ where } v \in A \text{ and } i \geq 0\}$ [marked as problem j in the solution]
- m) Recall that for a string w , $|w|$ denotes the length of w . $\Sigma = \{0,1\}$

$L1 = \{w \in \Sigma^* : w \text{ contains } 11\}$

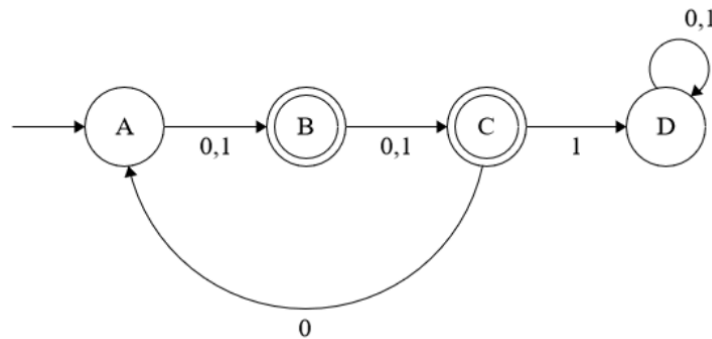
$L2 = \{x\#y : y \in L1, x \in \Sigma^*, |x| = |y|\}$

Construct a CFG for $L2$.

- n) $L = \{w \in \{0,1\}^* : w_1\#w_2 : \text{number of 0s } w_1 \text{ is equal to number of 1s in } w_2.\}$
- o) $L = \{w \in \{0,1\}^* : w_1\#w_2 : \text{length of } w_2 \text{ is double of length of } w_1.\}$

Construct a Context-free Grammar for the language expressed by the following Finite Automata:

- a) Convert the Regular Expression into a CFG: $((ab)^* + (a+b^3)cb)^*$
- b) Convert the Regular Expression into a CFG: $(a^*b^* + (ac+b^*c)b)^*$
- c) Convert the following DFA into a CFG:



Question 2: Derivation, Parse tree, Ambiguity

I.

$S \rightarrow ASB \mid SS \mid SAS \mid A$

$A \rightarrow ASS \mid BS \mid B$

$B \rightarrow 00 \mid 11 \mid 01 \mid 1$

Given the above Context-Free Grammar, answer the following questions:

- a) Show the Leftmost Derivation of 00010111
- b) Sketch the parse tree corresponding to the derivation you gave in (a).
- c) Show the Rightmost Derivation of 00010111
- d) Sketch the parse tree corresponding to the derivation you gave in (c).

II. $S \rightarrow 0S1 \mid 1S0 \mid SS \mid 01 \mid 10$

Given the above Context-Free Grammar, answer the following questions:

- a) Show that the grammar above is ambiguous by demonstrating two different parse trees for 011010.
- b) Find a string w of length six such that w has exactly one parse tree in the grammar above.

Question 3: Pushdown Automata

- a) $L = \{ w \in \{0,1\}^* : w = 0^n 1^n, \text{ where } n \text{ is odd.} \}$
- b) $L = \{ w \in \{0,1\}^* : w = 0^m 1^n, \text{ where } m, n \geq 1 \text{ and } m \geq n. \}$
- c) $L = \{ w \in \{0,1\}^* : w \mid \text{the length of } w \text{ is divisible by four} \}$
- d) $L_1 = \{ w \in \{0,1\}^* : \text{the number of 1s in } w \text{ is multiple of 3.} \}$
 $L_2 = \{ w \in \{0,1\}^* : w \text{ contains even numbers of 0.} \}$
Construct a PDA for $L = \{ w \in \{0,1\}^* : w = uv, \text{ where } u \in L_1, v \in L_2 \text{ and } |u| = |v| \}$
- e) $L = \{ w \in \{0,1\}^* : w = 0^i 1^j 0^k, \text{ where } j = i+k \text{ and } i, k \geq 0. \}$
- f) $L = \{ w \in \{0,1\}^* : w = 0^i 1^j 0^k, \text{ where } i+j = k \text{ and } i, k \geq 0. \}$
- g) $L = \{ w \in \{0,1\}^* : w_1 \# w_2 : \text{number of 00 substrings in } w_1 \text{ is equal to number of 11 in } w_2. \}$

- h) $L = \{ w \overline{w}^R : w \in \{0,1\}^* \}$

$$L = \{ w \overline{w}^R : w \in \{0,1\}^* \}$$

Here, $\overline{w}^R$ denotes the reverse complement of the string w .

For example, $01001101 \in L$, because the second half of the string, 1101, is the reverse complement of the first half, 0100, i.e. $1101 = \overline{0100}^R$